

CHE 201 Fall 2019 Quiz 2

Joey Wilczynski

TOTAL POINTS

50 / 100

QUESTION 1

1 85 pts

1.1 a 35 / 40

- **0 pts** Correct

✓ - **5 pts** h2 is not calculated or wrong

- **10 pts** KE is not calculated

- **5 pts** h1 is wrong

- **10 pts** PE not calculated

1.2 b+c 0 / 45

- **0 pts** Correct

✓ - **45 pts** not done or wrong

- **5 pts** unit conversion

- **10 pts** b is wrong

QUESTION 2

2 2 15 / 15

✓ - **0 pts** Correct

- **15 pts** not done

NAME: Joey WICZYNSKI

UIN: _____

QUIZ 2: (100 pts) (Close book/closed notes) Show all your work for full credit

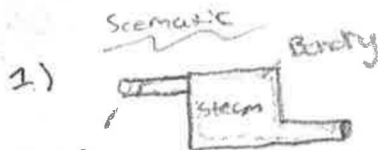
1. (85 points) The power output of steam turbine is 5MW, and the inlet and the exit conditions are as follow: $P_1=2\text{MPa}$, $T_1=400^\circ\text{C}$, $V_1=50\text{m/s}$, $z_1=10\text{m}$; $P_2=15\text{kPa}$, $x_2=0.90$, $V_2=180\text{m/s}$, $z_2=6\text{m}$. a) Compare the magnitudes of Δh , ΔKE , ΔPE . b) Determine the work done per unit mass of the steam flowing through the turbine. c) Calculate the mass flow rate of the steam.

2. (15 points) Briefly explain the process that Lanzatech patented. (Monday's guest speaker; Rachel Brenc)

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right) \quad \dot{m} = \frac{AV}{v}$$

$$1 \text{ joule} = 1 \text{ N}\cdot\text{m}$$

$$1 \text{ N} = 1 \text{ kg}\cdot\text{m/s}^2$$



$$P_1 = 2 \text{ MPa}$$

$$T_1 = 400^\circ\text{C}$$

$$V_1 = 50 \text{ m/s}$$

$$z_1 = 10 \text{ m}$$

$$P_2 = 15 \text{ kPa}$$

$$x_2 = 0.90$$

$$V_2 = 180 \text{ m/s}$$

$$z_2 = 6 \text{ m}$$

Engineering Model

- control volume

$$\dot{m}_i = \dot{m}_e = \dot{m}$$

- steady state

- 1 inlet 1 outlet

Energy Balance: $\dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right)$

Mass Balance: $\dot{m} = \frac{AV}{v}$

a) $(h_i - h_e) = ? \text{ kJ/kg}$

$$\left(\frac{V_i^2 - V_e^2}{2} \right)$$

$$(gz_i - gz_e)$$

$$10\text{m} \cdot 9.81 = 98.1 \text{ m}^2/\text{s}^2$$

$$(3247.6 - 2749.5)$$

$$0.1512$$

$$\Delta H = 498.1$$

$$\frac{50^2 - 180^2}{2} = \frac{-14950}{1000} = -14.95$$

b) $\dot{m}(h_i - h_e)$

$$3247.6 - 2749.5$$

$$v = (1 - 0.90) \cdot v_g + v_f(0.90)$$

#2 ON BACK
↓

2.) The process that was discussed in class was how her company is taking waste gases/fuel and using a micro organism to break it down, then it gets converted to ethanol to be used for in making things such as Jet fuel
clean reusable energy.